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Optimal Production Planning in a Multi-Product Stochastic Manufacturing System with Long-Run Average Cost *

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Abstract. This paper is concerned with the problem of production planning in a flexible manufacturing system consisting of a single or parallel failure-prone machines producing a number of different products. The objective is to choose the rates of production of the various products over time in order to meet their demands at the minimum long-run average cost of production and surplus. The analysis proceeds with a study of the corresponding problem with a discounted cost. It is shown using the vanishing discount approach for the average cost problem that the Hamilton-Jacobi-Bellman equation in terms of directional derivatives has a solution consisting of the minimal average cost and the so-called potential function. The result helps in establishing a verification theorem, and in specifying an optimal control policy in terms of the potential function. The results settle a hitherto open problem as well as generalize known results.

Keywords: Production planning, stochastic dynamic programming, vanishing discount approach, optimal control, long-run average cost.

1. Introduction

Beginning with Bielecki and Kumar (1988), there has been a considerable interest in studying the problem of production planning in stochastic manufacturing systems with the objective of minimizing long-run average cost. Bielecki and Kumar (1988) dealt with a single machine (with two states: up and down), single product problem with linear holding and backlog costs. Because of the simple structure of their problem, they were able to obtain an explicit solution for the problem, and thus verify the optimality of the resulting policy. They showed that the so-called hedging point policy is optimal in their simple model. It is a policy to produce at full capacity if the surplus is smaller than a threshold level, produce

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nothing if the surplus is higher than the threshold level, and produce as much as possible but no more than the demand rate when the surplus is at the threshold level.

Sharifnia (1988) dealt with an extension of the Bielecki-Kumar model with more than two machine states. Liberopoulous and Caramanis (1998) showed that Sharifnia's method for evaluating hedging point policies applies even when the transition rates of the machine states depend on the production rate. Liberopoulous and Hu (1995) obtained monotonicity of the threshold levels corresponding to different machine states. All of these papers, however, are *heuristic* in nature, since they do not rigorously prove the optimality of the policies for their extensions of the Bielecki-Kumar Model.

The difficulty in proving the optimality lies in the fact that when the problem is generalized to include convex costs and multiple machine capacity levels, explicit solutions are no longer possible, whereas the Bielecki-Kumar proof of optimality depended on being able to explicitly obtain the value function of the problem. One then needed to develop appropriate dynamic programming equations, existence of their solutions, and verification theorems for optimality. This was accomplished by Sethi et al. (1997), referred to as SSTZ hereafter. They used the vanishing discount approach to prove the optimality of hedging point policy for convex surplus cost and linear or convex production cost. Their results make precise the heuristic treatments of several manufacturing system problems carried out by Sharifnia (1988), Liberopoulous and Hu (1995), and others. Also, the results extend the restricted class of *stable* controls considered in Bielecki and Kumar (1988) to the natural class of admissible controls.

But SSTZ only considered a single product problem, and the multi-product problem continued to remain open. Once again there has been a number of heuristic analyses of the multi-product problem. Srivatsan (1993) and Srisvatsan and Dallery (1998) have considered a two-product problem. Caramanis and Sharifnia (1991) decompose a multi-product problem into analytically tractable single-product problem in order to obtain near-optimal hedging points for the problem. Other existing results for multi-product systems involve either approximations or numerical solutions (Veatch, 1992; Yan, Yin and Lou, 1994; Wein, 1991; Liberopoulous, 1992).

The problem has remained open because of possible nondifferentiability of the value function, particularly on switching manifolds. Ghosh, Arapostathis, and Marcus (1997) deal with the difficulty by approximating the problem with the incorporation a *nondegenerate* diffusion term into the dynamics of the multi-product manufacturing system. This allowed them to rigorously analyze the resulting average cost minimization problem. They were able to obtain a sufficiently smooth solution of the dynamic programming equation in the presence of the diffusion process, and thus verify the solution to be the value function for their problem.

However, in the absence of a diffusion term, the analysis of Ghosh et al. (1997) does not apply. Moreover, single-product model analysis of SSTZ does not apply to the model under consideration since the value function in SSTZ turns out to be continuously differentiable, and this issue still remains open in the multiproduct case.

For our treatment of the multi-product problems, we begin with the results for the corresponding discounted cost problem, which can be obtained by using the methodology developed by Presman, Sethi and Zhang (1995) for optimal control of flowshops. Their methodology utilizes the existence of optimal piecewise-deterministic controls available for systems under consideration. Then they are able to provide a verification theorem and the existence of an optimal feedback control based on the Hamilton-Jacobi-Bellman equation in terms of directional derivatives (HJBDD). Since multi-machines flowshops with state constraints are more complicated than the discounted cost multi-product problems, we simply state their results in an analogous manner as they pertain to our problem. Then we use a vanishing discount approach to address the average cost problem under consideration.

Two major contributions are made in order to implement the vanishing discount approach. The first one (Theorem 2) constructs a control policy which takes any given system state to any other state in a time whose r-th moment ($\forall r \geq 1$) has a finite expectation. The second one (Theorems 4 and 5) obtains a solution of the HJBDD equation by using convex analysis and properties of viscosity solutions and directional derivatives. These results and others derived in the paper allow us to prove the verification theorem (Theorem 6).

In this paper, we deal with a single or parallel machine manufacturing system with convex cost of holding and backloging and without diffusion. The plan of the paper is as follows. In Section 2 we introduce the problem and specify required assumptions. Section 3 is devoted to the study of the value function associated with the corresponding control problem with a discounted cost. It establishes the results needed for the so-called *vanishing discount approach* often used for analysis of average cost minimization problems. In particular, we derive a key lemma showing that one can go from any point in the multidimensional state space to any other point in *finite* expected time. In Section 4 the Hamilton-Jacobi-Bellman (HJB) equation and the HJBDD equations are specified for the average cost problem and a verification theorem for optimality over the class of admissible controls is given. Moreover, by using the vanishing discount approach, it is shown that the HJB equation has a viscosity solution, which turns out in this case to be also a classical solution of the HJBDD equation. The analysis helps us in §6 to specify the optimal control policy for the average cost problem under consideration. Section 5 concludes the paper.

2. Problem Formulation

We consider a multi-product manufacturing system with stochastic production capacity and constant demand for each product over time. In order to specify the model, let $x(t)$, $u(t)$, and d denote the surplus level (the state variable), the production rate (the control variable), and the constant demand rate, respectively. We assume $x(t) = (x_i(t)) \in \mathbb{R}^n$, $u(t) = (u_i(t)) \in \mathbb{R}^n_+$, $t \geq 0$, and $d = (d_i) \in \mathbb{R}^n_+$ is a constant vector.

The system equation is

$$\dot{x}(t) = u(t) - d, \; x(0) = x. \tag{1}$$

Let $\alpha(\cdot)$ denote a Markov process with a finite state space $\mathcal{M} = \{0, 1, 2, \ldots, m\}$, with $\alpha(t)$ representing the production capacity of the system at time t. This representation for $\mathcal{M}$ stands usually, but not necessarily, for the case of m identical machines, each with a unit capacity and having two states: up and down. This is not an essential assumption. In general, $\mathcal{M}$ could be any finite set of nonnegative numbers representing production capacities in the various states of the system.

Definition 1. A production control process $u(\cdot) = \{u(t),\, t \geq 0\}$ is admissible if (i) $u(t) \in \mathcal{F}_t \equiv \sigma(\alpha(s), 0 \leq s \leq t)$ and (ii) $u_1(t) + u_2(t) + \cdots u_n(t) \leq \alpha(t),\ u_i(t) \geq 0,\ i = 1, \ldots, n,\ t \geq 0$.

Let

$$\Omega(k) = \{u:\ u_i \geq 0,\ i = 1, \ldots, n,\ \sum_{i=1}^{n} u_i \leq k\}$$

denote the set of feasible controls given the machine capacity k.

Remark. Before proceeding further, we should mention that the model we have formulated is standard in the literature. See Chapter 9 of Gershwin (1994) and Remarks 2.2 and 2.3 in Chapter 2 of Sethi and Zhang (1994) for details and references devoted to the formulation and justification of the model.

We denote by $\mathcal{A}(k)$ the collection of admissible controls with the initial condition $\alpha(0) = k$.

Definition 2. A function $w(x,k)$ defined on $\mathbb{R}^n \times \mathcal{M}$ is called an admissible feedback control, or simply a feedback control, if (i) for any given initial surplus x and production capacity k, the equation

$$\dot{x}(t) = w(x(t), \alpha(t)) - d$$

has a unique solution and (ii) the control defined by $u(\cdot) = \{u(t) = w(x(t), \alpha(t)), t \geq 0\} \in \mathcal{A}(k)$. With a slight abuse of notation, we simply call $w(x,k)$ a feedback control when no ambiguity arises.

Let $h(\cdot) : \mathbb{R}^n \mapsto \mathbb{R}_+$ denote the surplus (inventory/backlog) cost and production cost, respectively. For any $u(\cdot) \in \mathcal{A}(k)$, define

$$J(x, k, u(\cdot)) = \limsup_{T \to \infty} \frac{1}{T} E \int_0^T h(x(t), u(t))\, dt, \tag{2}$$

where $x(\cdot)$ is the surplus process corresponding to the production process $u(\cdot)$. Our goal is to choose $u(\cdot) \in \mathcal{A}(k)$ so as to minimize the cost functional $J(x, k, u(\cdot))$.

We assume the cost function $h(\cdot)$ and the production capacity process $\alpha(\cdot)$ to satisfy the following:

(A1) $h(\cdot,\cdot)$ defined on $\mathbb{R}^n \times \Omega(m)$, is a nonnegative convex function with $h(0,0) = 0$. There are positive constants C_1, C_2, and $\kappa_0 \geq 1$ such that for any fixed u,

$$h(x,u) \geq C_1 |x|^{\kappa_0} - C_2, \qquad x \in \mathbb{R}^n;$$

moreover, there are constants $C_1' > 0$ and $\kappa \geq \kappa_0$ such that

$$|h(x,u) - h(y,u')| \leq C_1'(1 + |x|^{\kappa-1} + |y|^{\kappa-1})(|x-y| + |u-u'|) \quad \forall x, y \in \mathbb{R}^n, u, u' \in \Omega(m).$$

(A2) $\alpha(\cdot)$ is a finite state Markov chain with generator Q, where $Q = (q_{ij})$, $i, j \in \mathcal{M}$ is a $(m+1) \times (m+1)$ matrix such that $q_{ij} \geq 0$ for $i \neq j$ and $q_{ii} = -\sum_{i \neq j} q_{ij}$. We assume that Q is *strongly irreducible* in the following sense: the equations

$$\nu Q = 0 \quad \text{and} \quad \sum_{i=0}^{m} \nu_i = 1$$

have a unique solution $\nu = (\nu_0, \nu_1, \ldots, \nu_m)$ with $\nu_k > 0,\ k = 0, 1, \ldots, m$. The vector ν is called the *equilibrium distribution vector* of the Markov chain $\alpha(\cdot)$.

(A3) The average capacity $\bar{\alpha} \equiv \sum_{i=0}^{m} i\nu_i > d$.

Remark. Assumption (A1) is the usual assumption on the growth rate of the surplus cost function to ensure the existence of solutions to the HJB equation of the optimal control problem. Assumption (A3)–the average capacity of the machine is greater than demand—is necessary in the sense that if this were not true, then even if the system is always producing at its maximum capacity, it still would not meet the demand and the backlog would build up without bound over time with the consequence that no optimal solution with a finite cost would exist.

In (2), we have defined our cost function to be the limit superior ($\limsup$) of the finite horizon average costs as the horizon increases, and our optimal control problem to be one of minimizing the cost over the (natural) class of admissible controls. Since the limit of the finite horizon costs may not exist for all controls in this admissible class, it may be of interest to know the class of controls over which the optimal control also minimizes the limit inferior ($\liminf$) of the finite horizon costs. For this purpose, we define a smaller class of controls as follows: A control $u(\cdot) \in \mathcal{A}(k)$ is called *stable* if the condition

$$\lim_{T \to \infty} \frac{E|x(T)|^{\kappa+1}}{T} = 0 \tag{3}$$

holds, where $x(\cdot)$ is the surplus process corresponding to the control $u(\cdot)$ with $(x(0), \alpha(0)) = (x, k)$ and κ is defined in Assumption $(A1)$. Condition (3) allows us to use Dynkin's formula to obtain the required verification theorem established in Section 4.

Let $\mathcal{B}(k) \subset \mathcal{A}(k)$ denote the class of stable controls. It can be seen in the next section that the set of stable admissible controls $\mathcal{B}(k)$ is nonempty.

We will show in Section 4 that there exists a constant λ^*, independent of the initial condition $(x(0), \alpha(0)) = (x, k)$, and a stable Markov control policy $u^*(\cdot) \in \mathcal{A}(k)$ such that $u^*(\cdot)$ is optimal, i.e., it minimizes the cost defined by (2) over all $u(\cdot) \in \mathcal{A}(k)$, and furthermore,

$$\lim_{T \to \infty} \frac{1}{T} E \int_0^T h(x^*(t), u^*(t))\, dt = \lambda^*,$$

where $x^*(\cdot)$ is the surplus process corresponding to $u^*(\cdot)$ with $(x(0), \alpha(0)) = (x, k)$. Moreover, for any other (stable) control $u(\cdot) \in \mathcal{B}(k)$,

$$\liminf_{T\to\infty} \frac{1}{T} E \int_0^T h(x(t),u(t))\,dt \geq \lambda^*.$$

Since we will use the vanishing discount approach to study our problem, we provide an analysis of the discounted problem in the next section.

3. Analysis of the Discounted Cost Problem

In order to derive the HJB equation for the average cost control problem formulated above and study the existence of its solutions, we introduce a corresponding control problem with the cost discounted at a rate $\rho > 0$. For $u(\cdot) \in \mathcal{A}(k)$, we define the expected discounted cost as

$$J^\rho(x,k,u(\cdot)) = E \int_0^\infty e^{-\rho t} h(x(t),u(t))\,dt.$$

Define the value function of the discounted problem as

$$V^\rho(x,k) = \inf_{u(\cdot)\in\mathcal{A}(k)} J^\rho(x,k,u(\cdot)). \tag{4}$$

The following result can be easily derived in a manner similar to Theorem 2 in Presman, Sethi and Zhang (1995), repeated also in Chapter 4 of Sethi and Zhang (1994); see also Presman, Sethi and Suo (1997).

LEMMA 1 *The value function $V^\rho(\cdot,\cdot)$ is convex, and satisfies the following growth condition*

$$|V^\rho(x,k) - V^\rho(x',k)| \leq C(1+|x|^\kappa + |x'|^\kappa)|x-x'| \tag{5}$$

for some positive constant C and all $x, x' \in \mathbb{R}^n, k \in \mathcal{M}$.

It can be easily shown that if the value function $V^\rho(\cdot,k)$ is C^1, then it satisfies the HJB equation associated with this problem

$$\rho V^\rho(x,k) = \inf_{u\in\Omega(k)} \{\langle V_x^\rho, u-d\rangle + h(x,u)\} + QV^\rho(x,\cdot)(k), \tag{6}$$

where $V_x^\rho(\cdot,k)$ is the partial derivative of the value function $V^\rho(\cdot,k)$ with respect to the state variable x, and $\langle\cdot,\cdot\rangle$ denotes the inner product

$$\langle r, u-d\rangle = \sum_{i=1}^n r_i(u_i - d_i).$$

Without requiring that the value function $V^\rho(\cdot,k)$ is C^1, we need to use a different approach. Since $V^\rho(\cdot,k)$ is convex, it is convenient to write an HJB equation in terms of *directional derivatives* (HJBDD). So, we first give the notion of these derivatives and some related properties of convex functions.

A function $f(x), x \in \mathbb{R}^n$, is said to have a *directional derivative* $\partial_p f(x)$ along direction $p \in \mathbb{R}^n$, defined by

$$\lim_{\varepsilon \downarrow 0} \frac{f(x+\varepsilon p) - f(x)}{\varepsilon} = \partial_p f(x),$$

whenever the limit exists. If a function $f(x)$ is differentiable at x , then $\partial_p f(x)$ exists for every p and

$$\partial_p f(x) = \langle \nabla f(x), p \rangle,$$

where $\nabla f(x)$ is the gradient of $f(x)$ at x. It is well known that a continuous convex function defined on a convex domain $\sum$ is differentiable almost everywhere, and has a directional derivative along any direction at any inner point of $\sum$ and along any admissible direction (i.e., such direction p that $x + \varepsilon p \in \sum$ for some $\varepsilon > 0$) at any boundary point of $\sum$.

Formally we can write the HJBDD for our problem as:

$$\rho v(x,k) = \inf_{u \in \Omega(k)} \{\partial_{u-d} v(x,k) + h(x,u)\} + Qv(x,k). \tag{7}$$

The following theorem can be proved by using the methodology developed in Presman, Sethi and Zhang (1995), which is based on the theory of piecewise deterministic processes.

THEOREM 1 (VERIFICATION) *(a) The value function $V^\rho(x,k)$ satisfies equation* (7) *for all $x \in I\!R^n$.*

(b) If some continuous convex function $\tilde{V}^\rho(x,k)$ satisfies (7) *and the growth condition* (5) *with $x' = 0$, then $\tilde{V}^\rho(x,k) \le V^\rho(x,k)$. Moreover, if there exists a feedback control $u(x,k)$ providing the infimum in* (7) *for $\tilde{V}^\rho(x,k)$, then $\tilde{V}^\rho(x,k) = V^\rho(x,k)$ and $u(x,k)$ is an optimal feedback control.*

(c) Assume that $h(x,u)$ is strictly convex in u for each fixed x and k. Let $u^*(x,k)$ denote the minimizer function of the right-hand side of (7)*. Then,*

$$\dot{x}(t) = u^*(x(t), \alpha(t)),\ x(0) = x$$

has a solution $x^*(t)$ and $u^*(t) = u^*(x(t), \alpha(t))$ is the optimal control.

According to (7), optimal feedback control policies are obtained in terms of the directional derivatives of the value function. Note that the uniqueness of the optimal control follows directly from the strict convexity of function $h(\cdot,\cdot)$ in either x or u or both and the fact that any convex combination of admissible controls for any given x is admissible.

In order to study the long-run average cost control problem using the *vanishing discount approach*, we must first obtain some estimates for the value function $V^\rho(x,k)$.

LEMMA 2 Let $\sigma = \inf\{t :\ \int_0^t (\alpha(s) - D)\, ds = l\}$ for any $l > 0$. *Then for any $r \ge 1$, there is a constant C independent of l such that $E\sigma^r \le C(l^r + 1)$.*

Proof. See SSTZ for a proof. □

LEMMA 3 *For any $(x,k) \in I\!R^n \times \mathcal{M}$, $y \in I\!R^n$, there exists a control policy $u(t), t \ge 0$, such that for $r \ge 1$,*

$$E\tau_0^r \le C(1 + \sum_{i=1}^n |y_i - x_i|^r),$$

where

$$\tau_0 \equiv \inf\{t \geq 0 : x(t) = y\}$$

and $x(t), t \geq 0$*, is the surplus process corresponding to the control policy* $u(t)$ *and initial condition* $(x(0), \alpha(0)) = (x, k)$.

Proof. We first show that the lemma is true for $y \geq x$, i.e., $y_i \geq x_i, i = 1, \ldots, n$. For $y \geq x$, let us define a feedback control policy by

$$u_j(z,k) = \begin{cases} \dfrac{(y_j - z_j)k}{\sum_{i=1}^n (y_i - z_i)} & \text{if } z \leq y, z \neq y, k < \displaystyle\sum_{i=1}^n d_i, \\ \dfrac{(k - \sum_{i=1}^n d_i)(y_j - z_j)}{\sum_{i=1}^n (y_i - z_i)} + d_j & \text{if } z \leq y, z \neq y, k \geq \displaystyle\sum_{i=1}^n d_i, \\ 0 & \text{otherwise.} \end{cases} \tag{8}$$

Then it is easy to see that when $z \leq y, z \neq y$,

$$\sum_{j=1}^n u_j(z,k) = k, \quad u(z,k) \geq 0, \tag{9}$$

and thus $u(\cdot,\cdot)$ defines a feedback control policy. Moreover, note that for all j such that $y_j > z_j$, the ratio

$$\frac{u_j(z,k) - d_j}{y_j - z_j} = \frac{k - \sum_{i=1}^n d_i}{\sum_{i=1}^n (y_i - z_i)}$$

is a constant for all k, and therefore when the machine is in the *up* state, defined by $\alpha(t) \geq \sum_{i=1}^n d_i$, the corresponding surplus process moves along the straight line joining y and z.

Now let us consider

$$X(t) = \sum_{i=1}^n x_i(t),$$

and let $X = \sum_{i=1}^n x_i$, $Y = \sum_{i=1}^n y_i$, and $D = \sum_{i=1}^n d_i$. Define

$$\theta \equiv \inf\{t \geq 0, X(t) = Y\}.$$

It can be shown that $\theta = \tau_0$. Note that under the defined feedback control policy $u(\cdot,\cdot)$, we have, for $t \leq \theta$,

$$\dot{X}(t) = \alpha(t) - D.$$

Thus by applying Lemma A.1 of SSTZ and Assumption (A3), we can conclude

$$E\theta^r \;\leq\; C_1(1 + |Y - X|^r)). \tag{10}$$

We now show that Lemma 3 is true for any x and y. Let

$$i_0 \in \operatorname*{argmin}_i \left\{ \frac{y_i - x_i}{d_i} \right\}.$$

If $y_{i_0} - x_{i_0} \geq 0$, then it reduces to the case we have just proved. Thus we may assume

$$y_{i_0} - x_{i_0} < 0,$$

and without loss of generality, we assume $i_0 = 1$.

Define

$$u^0(t) = 0, \quad 0 \leq t \leq \frac{|y_1 - x_1|}{d_1} \equiv t_0,$$

then the corresponding trajectory is

$$\begin{aligned} x_1(t_0) &= y_1, \\ x_i(t_0) &= x_i - \frac{x_1 - y_1}{d_1} d_i \\ &\leq x_i - \frac{x_i - y_i}{d_i} d_i = y_i. \end{aligned}$$

Define $\tilde{x} = x(t_0)$. Then as have shown before, there is a (feedback) control $u(\cdot, \cdot)$ such that

$$E\tilde{\tau}_0^r \leq C(1 + \sum_{i=1}^{n} |\tilde{x}_i - y_i|^r),$$

where

$$\tilde{\tau}_0 \equiv \inf\{t \geq 0 : x(t_0 + t) = y\}.$$

Define

$$u(t) = \begin{cases} 0 & 0 \leq t \leq t_0, \\ u(x(t), \alpha(t)) & t > t_0, \end{cases}$$

then we have

$$\begin{aligned} E\tau_0^r &= E(t_0 + \tilde{\tau}_0)^r \leq C_1(t_0^r + E\tilde{\tau}_0^r) \\ &\leq C(1 + \sum_{i=1}^{n} |y_i - x_i|^r). \end{aligned}$$

The lemma is thus proved. □

From the proof of the lemma we can state the following corollary.

COROLLARY 1 *Let* $x(\cdot)$ *be the surplus process corresponding to the feedback control constructed in* (8), *and let*

$$X(t) = \sum_{i=1}^{n} x_i(t).$$

If $x(0) = x \le y$*, then we can conclude that* $\tau_0 = \theta$*, where*

$$\tau_0 \equiv \inf\{t : x(t) = y\},$$
$$\theta \equiv \inf\{t : X(t) = \sum_{i=1}^{n} y_i\}.$$

Now we are in the position to prove the following theorem.

THEOREM 2 *For any* $(x, k), (y, k') \in I\!R^n \times \mathcal{M}$*, there exists a control policy* $u(\cdot)$ *such that for* $r \ge 1$*,*

$$E\tau_0^r \le C(1 + \sum_{i=1}^{n} |y_i - x_i|^r),$$

where

$$\tau \equiv \inf\{t \ge 0 : (x(t), \alpha(t) = (y, k')\}$$

and $x(\cdot)$ *is the surplus process corresponding to the control policy* $u(\cdot)$ *and initial condition* $(x(0), \alpha(0)) = (x, k)$*.*

Proof. We show that given the results in Lemma 3, this theorem can be proved similarly as in Lemma 3.3 of SSTZ.

We first assume $x \le y$. Take $R > 0$ to be large enough such that when $t \ge R$,

$$P(\alpha(t) = k|\alpha(0) = i) \ge \frac{\nu_k}{2}, \quad \forall i, k, \tag{11}$$

where $\{\nu_k : k = 0, 1, \cdots, m\}$ is the stationary distribution of the Markov chain $\alpha(\cdot)$. Let N be a constant such that $N > RD$.

We can construct a control policy through the following: Apply the control policy we constructed in (8) in the proof of Lemma 3, with y being replaced by $y + Nd = (y_1 + Nd_1, \cdots, y_n + Nd_n)$. Lemma 3.3 implies that in a finite period of time τ_1, the surplus process $x(\cdot)$ corresponding to this control policy will reach the point $y + Nd$. Then we switch the control to $u(t) = 0$, and the surplus process will reach y in a period of time N. We can then use the control policy constructed in (8) again to push the surplus process to the state $y + Nd$ in a period of time τ_2. A control policy can therefore be constructed by repeating this procedure.

Define

$$\begin{aligned}
\tau_1 &\equiv \inf\{t > 0 : \ x(t) = y + Nd\}, \\
\sigma_1 &= \tau_1 + N, \\
\tau_n &= \inf\{t > \sigma_{n-1}, x(t) = y + Nd\}, \\
\sigma_n &= \tau_n + N, n \ge 2.
\end{aligned} \tag{12}$$

It is obvious by our construction that $x(\sigma_n) = x$, $n = 1, 2, \ldots$. Furthermore, by the strong Markov property of $\alpha(\cdot)$, we have

$$\begin{aligned} P(\tau > \sigma_n) &\leq P(\alpha(\sigma_n) \neq k, \alpha(\sigma_{n-1}) \neq k, \ldots, \alpha(\sigma_1) \neq k) \\ &= P(\alpha(\sigma_n) \neq k | \alpha(\sigma_{n-1}) \neq k) \cdots P(\alpha(\sigma_2) \neq \\ &\quad k | \alpha(\sigma_1) \neq k) P(\alpha(\sigma_1) \neq k). \end{aligned} \tag{13}$$

Note that $\sigma_i - \sigma_{i-1} \geq N/D > R$, $i = 1, 2 \ldots$. We set $\sigma_0 = 0$ and $\tau_0 = 0$. By (11) we have

$$\begin{aligned} &P(\alpha(\sigma_1) \neq k) \leq 1 - \delta, \\ &P(\alpha(\sigma_i) \neq k | \alpha(\sigma_{i-1}) \neq k) \leq 1 - \delta, \quad i = 2, 3, \ldots, \end{aligned}$$

where $\delta \equiv \min\{\nu_k/2 : \ k \in \mathcal{M}\}$. Then from (13), we have

$$P(\tau > \sigma_n) \leq (1 - \delta)^n. \tag{14}$$

Recall that $\sigma_n - \sigma_{n-1} = \tau_n - \sigma_{n-1} + N$ and

$$\sigma_n - \sigma_{n-1} = \inf \{t > 0 : x(t + \sigma_{n-1}) = y + Nd\}.$$

Note that $x(\sigma_{n-1}) = x$. Apply Lemma 2 for any $r \geq 1$ to obtain

$$E|\sigma_n - \sigma_{n-1}|^r \leq C_1(1 + |y + Nd - x|^r) \leq C_2(1 + |y - x|^r),$$

where $C_1, C_2 > 0$ are constants independent of x. Therefore,

$$\begin{aligned} E\tau^r &= r \int_0^\infty t^{r-1} P(\tau > t)\, dt \\ &= r \sum_{n=1}^\infty E \int_{\tau_{n-1}}^{\tau_n} t^{r-1} P(\tau > t)\, dt \\ &\leq r \sum_{n=1}^\infty E \int_{\tau_{n-1}}^{\tau_n} t^{r-1} P(\tau > \tau_{n-1})\, dt \\ &\leq r \sum_{n=1}^\infty E \int_{\tau_{n-1}}^{\tau_n} t^{r-1} P(\tau > \sigma_{n-2})\, dt \\ &\leq \sum_{n=1}^\infty E(\tau_n - \tau_{n-1})^r (1 - \delta)^{n-2} \\ &\leq C_2(1 + |y - x|^r) \sum_{n=1}^\infty (1 - \delta)^{n-1} \\ &\leq C(1 + |y - x|^r). \end{aligned}$$

For any general pair of x and y, i.e., not necessarily $x \geq y$, we can use the same technique as in second half of proof for Lemma 3. ∎

With Theorem 2 in hand, we can follow the corresponding proofs in SSTZ to prove the following results.

THEOREM 3 *(a) There exists a constant $\rho_0 > 0$ such that $\{\rho V^\rho(0,0) : 0 < \rho \le \rho_0\}$ is bounded.*

(b) The function

$$W^\rho(x,k) \equiv V^\rho(x,k) - V^\rho(0,0) \tag{15}$$

is convex in x. It is locally uniformly bounded, i.e., there exists a constant $C > 0$ such that

$$|V^\rho(x,k) - V^\rho(0,0)| \le C(1 + |x|^{\kappa+1}) \quad \forall (x,k) \in I\!R \times \mathcal{M}, \rho \ge 0.$$

(c) $W^\rho(x,k)$ is locally uniformly Lipschitz continuous in x with respect to $\rho > 0$, i.e., for any $X > 0$, there exists a constant $C > 0$, independent of ρ, such that

$$|W^\rho(x,k) - W^\rho(x',k)| \le C|x - x'|$$

for all $k \in \mathcal{M}$ and all $|x|, |x'| \le X$.

4. Verification Theorem

The HJB equation associated with the long-run average cost optimal control problem formulated in §2 takes the following form

$$\lambda = \inf_{u \in \Omega(k)} \{\langle W_x(x,k), u - d\rangle + h(x,u)\} + QW(x,\cdot)(k), \tag{16}$$

and the corresponding HJB equation in the directional derivative sense (HJBDD) can be written as

$$\lambda = \inf_{u \in \Omega(k)} \{\partial W_{u-d}(x,k) + h(x,u)\} + QW(x,\cdot)(k), \tag{17}$$

where λ is a constant and W is a real-valued function defined on $I\!R \times \mathcal{M}$.

Before we define the solutions to the equations (16) and (17), we first introduce some notation. Let $\mathcal{G}$ denote the family of real-valued functions $W(\cdot,\cdot)$ defined on $I\!R \times \mathcal{M}$ such that

(i) $W(\cdot,k)$ is convex;

(ii) $W(\cdot,k)$ has polynomial growth, i.e., there are constants $\kappa, C > 0$ such that

$$|W(x,k)| \le C(1 + |x|^{\kappa+1}) \qquad \forall x \in I\!R.$$

A solution to the HJB or HJBDD equation is a pair (λ, W) with λ a constant and $W \in \mathcal{G}$. The function W is called the *potential function* or the *relative value function* for the control problem, if λ is the minimum long-run average cost. The terminology for W is motivated by the relations (15) and (18).

We now try to construct a solution (λ, W) to the HJB equation (16) as defined below in (18). Since the HJB equation involves the partial derivative $W_x(\cdot,\cdot)$, which may not exist for the defined function $W(\cdot,\cdot)$, we must first interpret the solution of the HJB equation (16) in the *viscosity sense*. This is done in Theorem 4. We refer to Fleming and Soner (1992) for the definition of viscosity solutions and some related results.

THEOREM 4 For $(x,k) \in I\!R \times \mathcal{M}$, *the following limits exist:*

$$\lambda = \lim_{\rho \to 0} \rho V^\rho(x,k) \quad \textit{and} \quad V(x,k) = \lim_{\rho \to 0} W^\rho(x,k). \tag{18}$$

Furthermore, (λ, V) *is a viscosity solution to the HJB equation* (16).

Proof. The proof is similar to the proof of Theorem 4.1 in SSTZ. □

Before we move on, we give some results concerning convex functions. We begin with some definitions. For any function $f(\cdot) : I\!R \mapsto I\!R$, the *superdifferential* $D^+f(x)$ and the *subdifferential* $D^-f(x)$ of the function f at x are defined, respectively, as follows,

$$D^+f(x) = \left\{ r \in I\!R^n : \limsup_{h \to 0} \frac{f(x+h) - f(x) - hr}{|h|} \le 0 \right\},$$
$$D^-f(x) = \left\{ r \in I\!R^n : \liminf_{h \to 0} \frac{f(x+h) - f(x) - hr}{|h|} \ge 0 \right\}.$$

The following results can be found in Clarke (1983).

LEMMA 4 *For a convex function* $f : I\!R^n \mapsto I\!R$, *we have the following:*

(i) *The subdifferential* $D^-f(x)$ *is nonempty.*

(ii) f *is differentiable at* x *if and only if* $D^-f(x)$ *is a singleton. In this case,* $D^-f(x) = D^+f(x)$.

(iii) *If the function* f *is differentiable on* $I\!R^n$, *then it is continuously differentiable on* $I\!R^n$.

(iv) *For every* $x \in I\!R^n,\ p \in I\!R^n$,

$$\partial_p f(x) = \max_{r \in D^- f(x)} \langle r, p \rangle.$$

From the properties of viscosity solutions, we know that the HJB equation (16) becomes an equality at each point where $V(\cdot,k)$ is differentiable. In fact, let $D^+V(x,k)$ and $D^-V^\rho(x,k)$ denote the superdifferential and subdifferential of the function $V(\cdot,k)$ at the point x, respectively. We have from Lemma 4 (ii) that if $V(\cdot,k)$ is differentiable at x, then

$$D^+V(x,k) = D^-V(x,k) = V_x(x,k).$$

Since $V(\cdot,k)$ is a continuous convex function, it is differentiable almost everywhere. For any x, we can take a sequence $x_n \to x$ such that $V(\cdot,k)$ is differentiable at x_n for each n. Thus from Theorem 4 we have

$$\begin{aligned} \lambda &= \inf_{u \in \Omega(k)} \{ \langle V_x(x_n,k), u - d \rangle + h(x_n,u) \} + QV(x_n,\cdot)(k) \\ &\le \langle V_x(x_n,k), u - d \rangle + h(x_n,u) + QV(x_n,\cdot)(k), \quad \forall\, 0 \le u \le k. \end{aligned}$$

Taking $x_n \to x$ as $n \to \infty$, we obtain from the continuity of $V(\cdot,k)$ and $h(\cdot,u)$,

$$\lambda \le \langle r, u - d \rangle + h(x,u) + QV(x,\cdot)(k), \quad \forall\, r \in \Gamma(x), \tag{19}$$

where

$$\Gamma(x) = \{r = \lim_{n\to\infty} V_x(x_n, k) : V(\cdot, k) \text{ is differentiable at } x_n\}.$$

From the linearity of the right-hand side of (19) in r, we can extend the inequality to the convex hull $\text{co}\Gamma(x)$ of the set $\Gamma(x)$. Taking $r_n \in \text{co}\Gamma(x)$ such that $r_n \to r$, the extension can be made to the closure $\bar{\text{co}}\Gamma(x)$, which by Theorem 2.5.1 of Clarke (1983) equals the set $D^- V(x, k)$. Thus we have shown that

$$\lambda \le \inf_{u\in\Omega(k)} \{\langle r, u-d\rangle + h(x,u)\} + QV(x,\cdot)(k) \quad \forall\, r \in D^- V(x,k).$$

On the other hand, the viscosity solution property implies the opposite inequality, and hence

$$\lambda = \inf_{u\in\Omega(k)} \{\langle r, u-d\rangle + h(x,u)\} + QV(x,\cdot)(k) \quad \forall\, r \in D^- V(x,k). \tag{20}$$

We can now prove the following existence theorem. The proof uses convex analysis to show (22) meaning that there exists a u^* which attains infimum in (20) for every $r \in D^- V(x, k)$. Then a property of directional derivatives immediately yields (23). Finally, we show that u^* is also an infimum of the right-hand side of (23), i.e., (24) holds.

THEOREM 5 (λ, V) *defined in Theorem* 4 *is a solution to the HJBDD equation* (17).

Proof. From Theorem 6.2 of Rockafeller (1972), we know that the relative interior of a nonempty convex set, denoted by $\text{ri}D^- V(x, k)$, is nonempty. We can now show that for any $\hat{r} \in \text{ri}D^- V(x,k)$, if we let u^* be such that

$$\lambda = \langle \hat{r}, u^* - d\rangle + h(x, u^*) + QV(x,\cdot)(k), \tag{21}$$

then (21) still holds when we replace $\hat{r}$ by any $r \in D^- V(x,k)$. In fact, by Theorem 6.4 of Rockafellar (1972), for any $r \in D^- V(x,k), r \ne \hat{r}$, there exists an $r' \in D^- V(x,k)$ such that

$$\hat{r} = \beta r + (1-\beta) r', \quad 0 < \beta < 1.$$

From (21) we can write

$$\lambda = \beta[\langle r, u^* - d\rangle + h(x,u^*) + QV(x,\cdot)(k)] + \\ (1-\beta)[\langle r', u^* - d\rangle + h(x,u^*) + QV(x,\cdot)(k)].$$

From (20), we have

$$\lambda \le \langle r, u^* - d\rangle + h(x,u^*) + QV(x,\cdot) + QV(x,\cdot)(k).$$

If

$$\lambda < \langle r, u^* - d\rangle + h(x,u^*) + QV(x,\cdot)(k),$$

then we must have

$$\lambda > \langle r', u^* - d \rangle + h(x, u^*) + QV(x, \cdot)(k),$$

which contradicts (20), i.e.,

$$\lambda = \inf_{u \in \Omega(k)} \{ \langle r', u - d \rangle + h(x, u) \} + QV(x, \cdot)(k)$$

because $r' \in D^- V(x, k)$. From this and (20), we can conclude that

$$\begin{aligned} \lambda &= \inf_{u \in \Omega(k)} \{ \langle r, u - d \rangle + h(x, u) \} + QV(x, \cdot)(k) \\ &= \langle r, u^* - d \rangle + h(x, u^*) + QV(x, \cdot)(k), \qquad \forall\, r \in D^- V(x, k). \end{aligned} \tag{22}$$

Apply Lemma 4 (iv) to conclude that

$$\lambda = \partial_{u^* - d} V(x, k) + h(x, u^*) + QV(x, \cdot)(k). \tag{23}$$

It follows directly from (23) that

$$\lambda \geq \inf_{u \in \Omega(k)} \{ \partial_{u-d} V(x, k) + h(x, u) \} + QV(x, \cdot)(k).$$

On the other hand, Lemma 4 (iv) allows us for every u to find an $r_u \in D^- V(x, k)$ such that $\langle r_u, u - d \rangle = \partial_{u-d} V(x, k)$. Using this fact and (22), we obtain the opposite inequality

$$\begin{aligned} \langle r_u, u - d \rangle + h(x, u) + QV(x, \cdot)(k) &= \partial_{u-d} V(x, k) + h(x, u) + QV(x, \cdot)(k) \\ &\geq \langle r_u, u^* - d \rangle + h(x, u^*) + QV(x, \cdot)(k) \\ &= \lambda, \qquad u \in \Omega(k). \end{aligned}$$

Thus, we have shown that

$$\lambda = \inf_{u \in \Omega(k)} \{ \partial_{u-d} V(x, k) + h(x, u) \} + QV(x, \cdot)(k), \tag{24}$$

i.e., (λ, V) is a solution of the HJBDD equation (17). ∎

The following verification theorem can be proved now.

THEOREM 6 Let (λ, W) *be a solution to the* HJBDD *equation* (17). *Then*

(i) *If there is a control* $u^*(\cdot) \in \mathcal{A}(k)$ *such that*

$$\inf_{u \in \Omega(\alpha(t))} \{ \partial W_{u-d}(x^*(t), \alpha(t)) + h(x^*(t), u) \} = \partial W_{u^*(t)-d}(x^*(t), \alpha(t)) + h(x^*(t), u^*(t)) \tag{25}$$

for a.e. $t \geq 0$ *with probability* 1, *where* $x^*(\cdot)$ *is the surplus process corresponding to the control* $u^*(\cdot)$, *and*

$$\lim_{T \to \infty} \frac{EW(x^*(T), \alpha(T))}{T} = 0, \tag{26}$$

then

$$\lambda = J(x, k, u^*(\cdot)).$$

(ii) *For any* $u(\cdot) \in \mathcal{A}(k)$, *we have* $\lambda \leq J(x,k,u(\cdot))$, *i.e.,*

$$\limsup_{t\to\infty} E\int_0^t h(x(t),(u(t))\,dt \geq \lambda.$$

(iii) *Furthermore, for any (stable) control policy* $u(\cdot) \in \mathcal{B}(k)$, *we have*

$$\liminf_{t\to\infty} \frac{1}{t}E\int_0^t h(x(t),u(t))\,dt \geq \lambda. \tag{27}$$

Proof. We begin with the proof of (i). Since (λ, W) is a solution to the HJBDD equation (17) and $u^*(\cdot)$ satisfies the condition (24), we have

$$\partial W_{u^*(t)-d}(x^*(t),\alpha(t)) + QW(x^*(t),\cdot)(\alpha(t)) = \lambda - h(x^*(t),u^*(t)). \tag{28}$$

Since $W \in \mathcal{G}$, we can apply Dynkin's formula (see Fleming and Rishel (1975)) to get

$$\begin{aligned} EW(x^*(T),\alpha(T)) &= W(x,k) + E\int_0^T \big(\partial W_{u^*(t)-d}(x^*(t),\alpha(t)) + QW(x^*(t),\cdot)(\alpha(t))\big)\,dt \\ &= W(x,k) + E\int_0^T (\lambda - h(x^*(t),u^*(t)))\,dt \\ &= W(x,k) + \lambda T - E\int_0^T h(x^*(t),u^*(t))\,dt, \end{aligned} \tag{29}$$

where we have used the equation (28). We can rewrite (29) as

$$\lambda = \frac{1}{T}\big(EW(x^*(T),\alpha(T)) - W(x,k)\big) + \frac{1}{T}E\int_0^T h(x^*(t),u^*(t))\,dt,$$

and the first part of the theorem is proved by taking the limit as $T \to \infty$ and using the condition (26).

For the proof of (iii), if $u(\cdot) \in \mathcal{B}(k)$, then by $W \in \mathcal{G}$, we know that

$$\lim_{T\to\infty} \frac{EW(x(T),\alpha(T))}{T} = 0.$$

Moreover, from the HJBDD equation (17) we have

$$W_{u(t)-d}(x(t),\alpha(t)) + QW(x(t),\cdot)(\alpha(t)) \geq \lambda - h(x(t)) - c(u(t)).$$

Now (27) can be proved similarly as before.

Finally, we apply the Tauberian theorem (see Section 7 of Sethi et al., 1997 or Sznadjer and Filar, 1992) to show (ii), i.e., the optimality of the control $u^*(\cdot)$ among all admissible controls. Let $u(\cdot) \in \mathcal{A}(k)$ be any policy and $x(\cdot)$ be the corresponding surplus process. Suppose that

$$J(x,k,u(\cdot)) < \lambda. \tag{30}$$

Write

$$f(t) = Eh(x(t), u(t)).$$

Without loss of generality we may assume that

$$\int_0^t f(s)ds < \infty$$

for each $t > 0$, otherwise $J(x, k, u(\cdot)) = \infty$. Note that

$$J(x, k, u(\cdot)) = \limsup_{t\to\infty} \frac{1}{t} \int_0^t f(s)ds,$$

while

$$\rho J^\rho(x, k, u(\cdot)) = \rho \int_0^\infty e^{-\rho s} f(s)\, ds.$$

Therefore, we can apply the Tauberian theorem (see SSTZ Theorem 6.1) to obtain

$$\limsup_{\rho\to 0} \rho J^\rho(x, k, u(\cdot)) < \lambda. \tag{31}$$

On the other hand, we know from the Tauberian theorem that

$$\lim_{\rho\to 0} \rho V^\rho(x, k) = \lambda.$$

Therefore, the last equation and (31) imply the existence of a $\rho > 0$ such that

$$\rho J^\rho(x, k, u(\cdot)) < \rho V^\rho(x, k),$$

which contradicts the definition of $V^\rho(x, k)$. Thus (ii) is proved. □

5. Concluding Remarks

In this paper, we have developed the theory of dynamic programming in terms of directional derivatives for single or parallel machines, convex cost, multi-product stochastic manufacturing systems with the long-run average cost minimization criterion. We have generalized previous results as well as provided a theoretical framework for various heuristic analyses of such systems in the literature.

Further research should focus on extending the analysis to more complex manufacturing systems such as flowshops and jobshops. For such models with discounted cost criteria, see Sethi and Zhang (1994).

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